

House Price Prediction & Analysis

A Comparative Study of Linear Regression and Decision Tree Regressor

Prepared By	Amanullah Fazil S
Project Title	House Price Prediction & Analysis
Algorithms Used	Linear Regression Decision Tree Regressor
Dataset	California Housing Dataset

Objective

This project builds and compares predictive models to estimate California house prices using features such as median income, house age, average rooms, and location. Two modeling approaches — Linear Regression and a Decision Tree Regressor — are trained, tuned, and evaluated, with the goal of identifying which technique better captures the underlying patterns in the housing data and explaining why.

Tasks Completed

1	Load and explore the California Housing Dataset; check shape, info, describe, and null values.
2	Visualize feature correlations using a heatmap; identify the top 3 features most correlated with price.
3	Preprocess the data: handle missing values and scale numerical features using StandardScaler.
4	Split the data into 80% training and 20% testing sets.
5	Train a Linear Regression model and evaluate using MAE, MSE, and R ² score.
6	Train a Decision Tree Regressor; tune max_depth (3, 5, 10) and report the best results.
7	Plot Actual vs. Predicted values for both models.
8	Compare both models in a summary table and conclude which performed better and why.

Report

1. Data Loading & Exploration

The dataset contains 20,640 rows and 10 columns. Summary statistics generated with `describe()` — mean, standard deviation, min, max, and quartiles — confirmed that all numerical features fall within reasonable ranges. A null-value check showed that every column is complete except `total_bedrooms`, which has 207 missing values.

Sample rows from the California Housing Dataset

longitude	latitude	housing_median_age	total_rooms	total_bedrooms	population	households	median_income	median_house_value	ocean_proximity
-122.23	37.88	41.0	880.0	129.0	322.0	126.0	8.3252	452600.0	NEAR BAY
-122.22	37.86	21.0	7099.0	1106.0	2401.0	1138.0	8.3014	358500.0	NEAR BAY
-122.24	37.85	52.0	1467.0	190.0	496.0	177.0	7.2574	352100.0	NEAR BAY
-122.25	37.85	52.0	1274.0	235.0	558.0	219.0	5.6431	341300.0	NEAR BAY

2. Feature Correlation Analysis

A correlation heatmap was generated across all numerical features to understand how strongly each one relates to `median_house_value`. A focused bar chart was then produced to rank features purely by their correlation with the target variable.

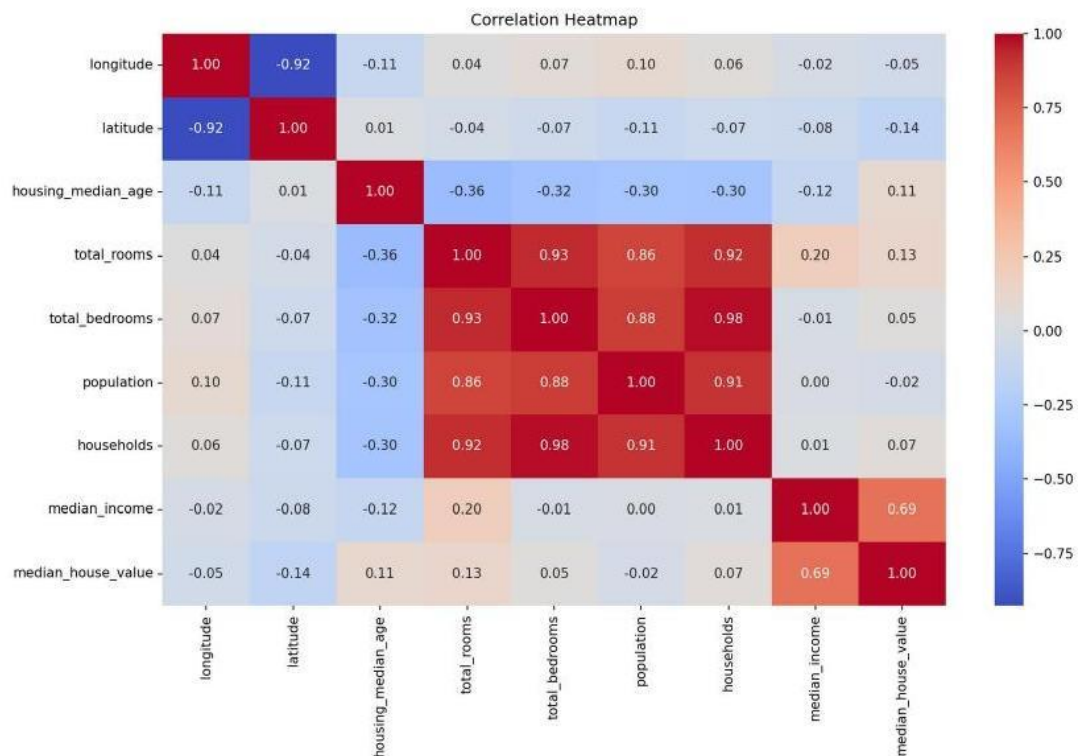
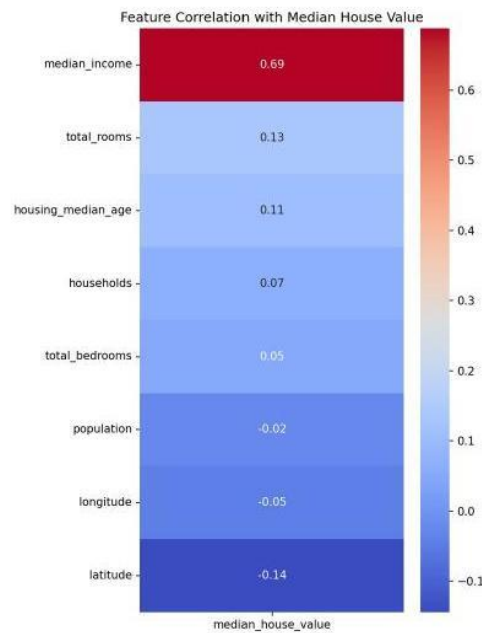


Figure 1. Correlation heatmap of all numerical features*Figure 2. Feature correlation ranked against median_house_value***Top 3 features most correlated with median_house_value:**

- median_income — correlation ≈ 0.69
- total_rooms — correlation ≈ 0.13
- housing_median_age — correlation ≈ 0.11

3. Preprocessing

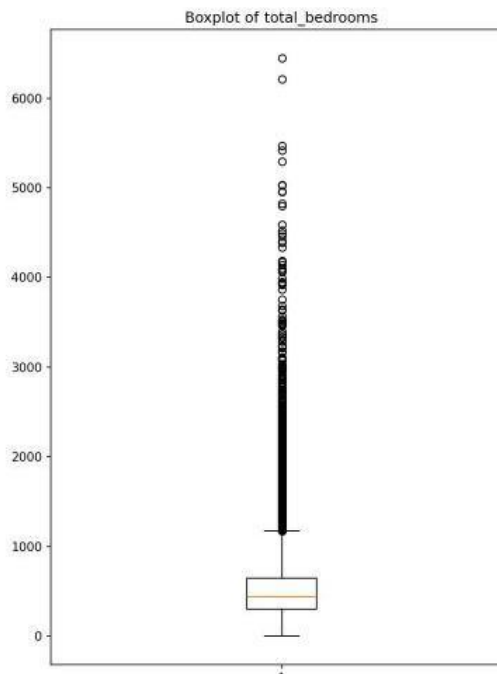


Figure 3. Boxplot of total_bedrooms showing a right-skewed, outlier-heavy distribution

The total_bedrooms column contained 207 missing values along with a large number of outliers, and showed no strong correlation with other features. Because the distribution is skewed and outlier-heavy, missing values were imputed using the median rather than the mean, which is more robust to extreme values. The categorical ocean_proximity column was one-hot encoded (drop_first=True), and all numerical features were standardized using StandardScaler so that every feature contributes on a comparable scale during model training.

4. Train–Test Split

The dataset was split using train_test_split with test_size = 0.2 and random_state = 42, producing 16,512 training rows and 4,128 testing rows across 12 engineered features (after one-hot encoding).

Set	Shape
Training Features	(16512, 12)
Testing Features	(4128, 12)
Training Labels	(16512,)
Testing Labels	(4128,)

5. Linear Regression Model

A Linear Regression model was trained on the scaled features. On the held-out test set, the model achieved a Mean Absolute Error (MAE) of \$50,670.74, a Mean Squared Error (MSE) of 4.91×10^9 , and an

R^2 score of 0.625. In other words, the model explains roughly 62.5% of the variance in house prices, with an average prediction error of about \$50,670 on unseen data.

6. Decision Tree Regressor

A Decision Tree Regressor was tuned using GridSearchCV with 5-fold cross-validation across max_depth values of 3, 5, and 10. The best cross-validated performance was achieved at max_depth = 10, which was carried forward for final evaluation.

On the test set, the tuned Decision Tree achieved an MAE of \$40,611.35, an MSE of 3.75×10^9 , and an R^2 score of 0.714 — clearly outperforming Linear Regression. The higher R^2 and lower error indicate that the Decision Tree captures non-linear relationships in the data that the linear model misses.

7. Actual vs. Predicted Values

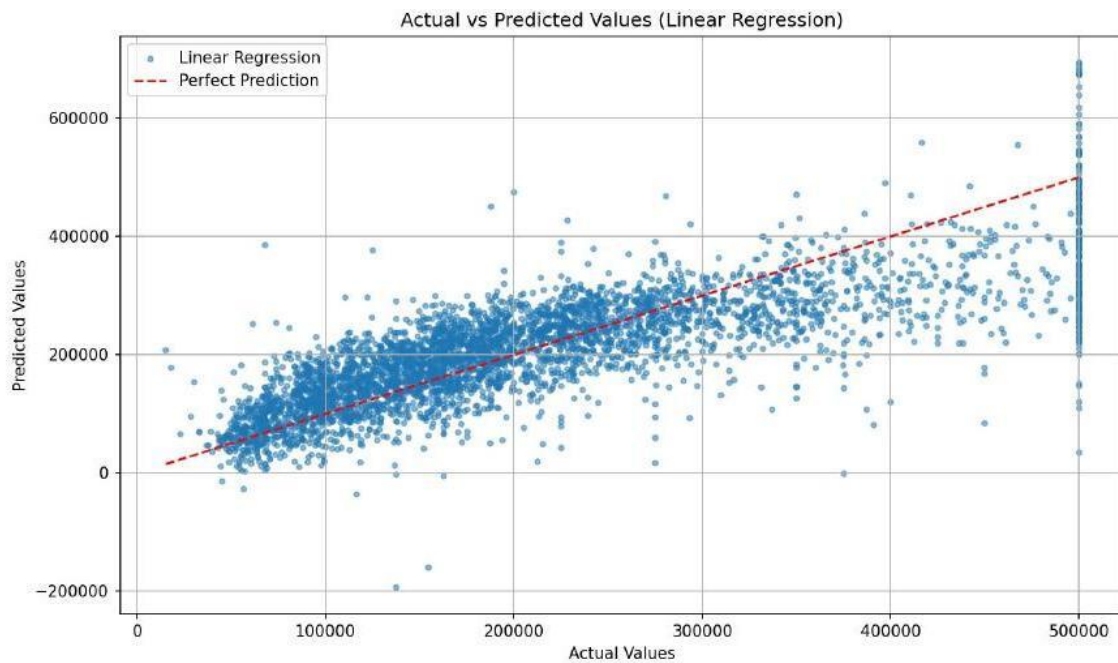


Figure 4. Actual vs. Predicted values — Linear Regression

The Linear Regression scatter plot shows predictions spread widely around the perfect-prediction line, particularly for higher-value homes. This reflects the model's limited ability to capture non-linear patterns, and is compounded by the fact that median_house_value is capped at \$500,001 in the source data.

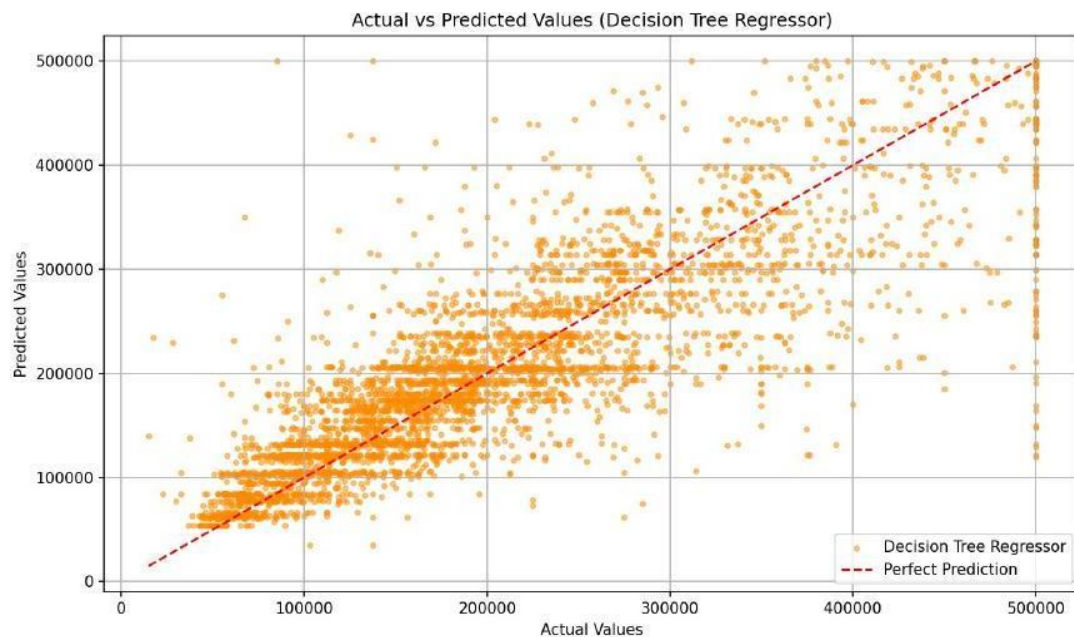


Figure 5. Actual vs. Predicted values — Decision Tree Regressor

The Decision Tree's predictions cluster more tightly around the perfect-prediction line across most of the price range, visually confirming its stronger fit relative to Linear Regression.

8. Model Comparison & Conclusion

Model	MAE (Train)	MAE (Test)	MSE (Train)	MSE (Test)	R ² (Train)	R ² (Test)
Linear Regression	49,594.84	50,670.74	4.68×10^9	4.91×10^9	0.6497	0.6254
Decision Tree Regressor	32,927.11	40,611.35	2.36×10^9	3.75×10^9	0.8234	0.7137

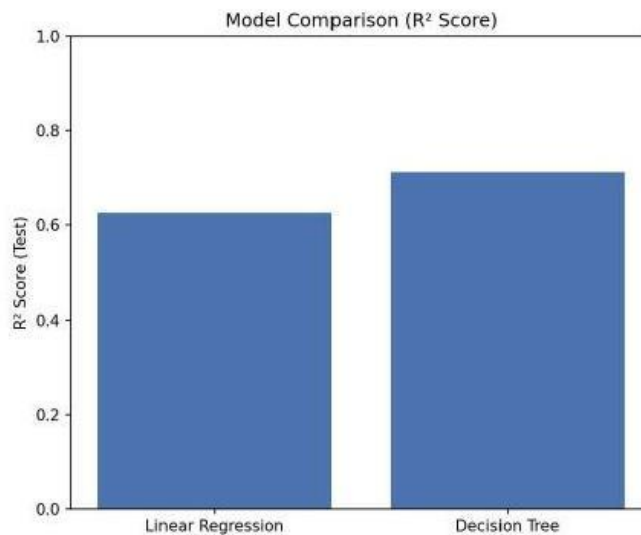


Figure 6. Test R² score comparison — Linear Regression vs. Decision Tree

Conclusion

The Decision Tree Regressor outperformed Linear Regression overall, achieving a test R² of 0.714 compared to 0.625, alongside a lower MAE (\$40,611 vs. \$50,671). Linear Regression underperformed because it assumes a strictly linear relationship between the input features and price, whereas the true relationships in the housing data — particularly around median_income, geographic location, and housing age — are non-linear. By recursively partitioning the feature space, the Decision Tree (tuned to a depth of 10) was able to capture these interactions and produce more accurate price estimates.

That said, the Decision Tree shows a noticeably larger gap between its training and test R² (0.823 vs. 0.714) than Linear Regression does, which points to a degree of overfitting to the training data. This is a known limitation of single decision trees at greater depth. Ensemble methods such as Random Forest or Gradient Boosting would likely reduce this overfitting while retaining the ability to model non-linear relationships, and represent a natural next step for improving generalization performance.

Key Takeaways

- median_income is by far the strongest predictor of house price (correlation ≈ 0.69).
- The Decision Tree Regressor (max_depth = 10) outperformed Linear Regression on every evaluation metric.
- Linear Regression's assumption of linearity limits its accuracy on this dataset.
- The Decision Tree shows mild overfitting; ensemble methods (Random Forest, Gradient Boosting) are recommended as a next step.